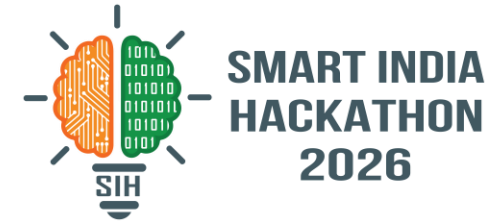
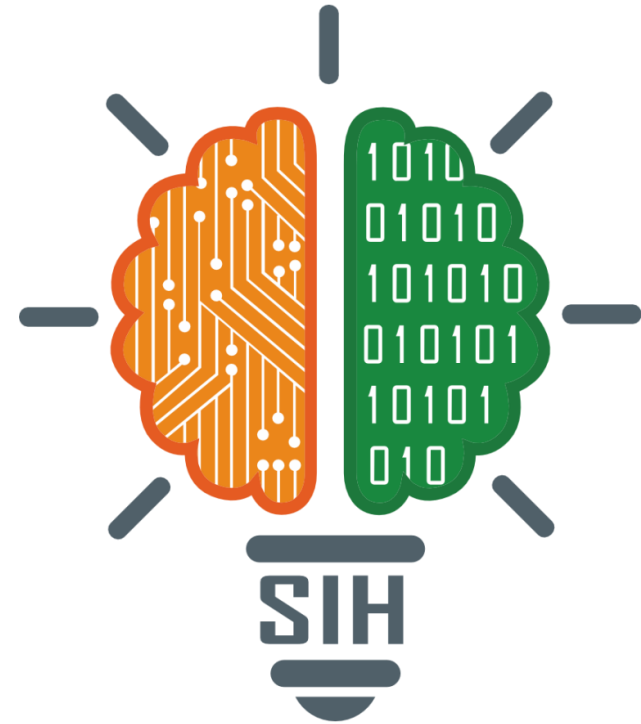


SMART INDIA HACKATHON 2026



- **Problem Statement ID –SIH26034**
- **Problem Statement Title-Software System to check compliance of Packaged Commodities under Legal Metrology(Packaged Commodities) Rules, 2011 by scanning products, images and labels.**
- **Theme-Miscellaneous**
- **PS Category- Software**
- **Team ID-151159**
- **Team Name –Govt2047**



PROBLEM OVERVIEW & PROPOSED SOLUTION

PROBLEM OVERVIEW

- **Manual Inspection** – Products are inspected individually by inspectors.
- **Time-Consuming** – Verifying all mandatory declarations requires significant time.
- **Human Errors** – Small, unclear, or missing violations may be overlooked.
- **High Product Volume** – Manual inspection becomes difficult at large scale.
- **Poor Record Tracking** – Maintaining, searching, and monitoring inspection records is challenging.

PROPOSED SOLUTION

- **AI-Powered Scanning** – Scans product images and detects relevant label areas.
- **OCR & Text Extraction** – Extracts key details such as MRP, net quantity, and dates.
- **NLP-Based Classification** – Identifies and categorizes extracted information.
- **Rule-Based Validation** – Validates details against applicable Legal Metrology rules.
- **Automated Reports & Dashboard** – Generates compliance reports and maintains evidence for easy tracking.

WHAT THE SYSTEM CHECKS



**Manufacturer /
Packer / Importer**



**Common / Generic
Name**



Net Quantity

Value + valid unit



MRP

Price + declaration context



Date / Best Before

Applicability + parsing



Consumer Care

Phone / email / address

KEY INNOVATION

AI extracts & interprets → deterministic
rules validate → human reviews →
evidence is retained.

TECHNICAL APPROACH

TECH STACK COMPONENTS

UI Web / Mobile
Responsive inspection UI

API Backend / API
FastAPI / Python services

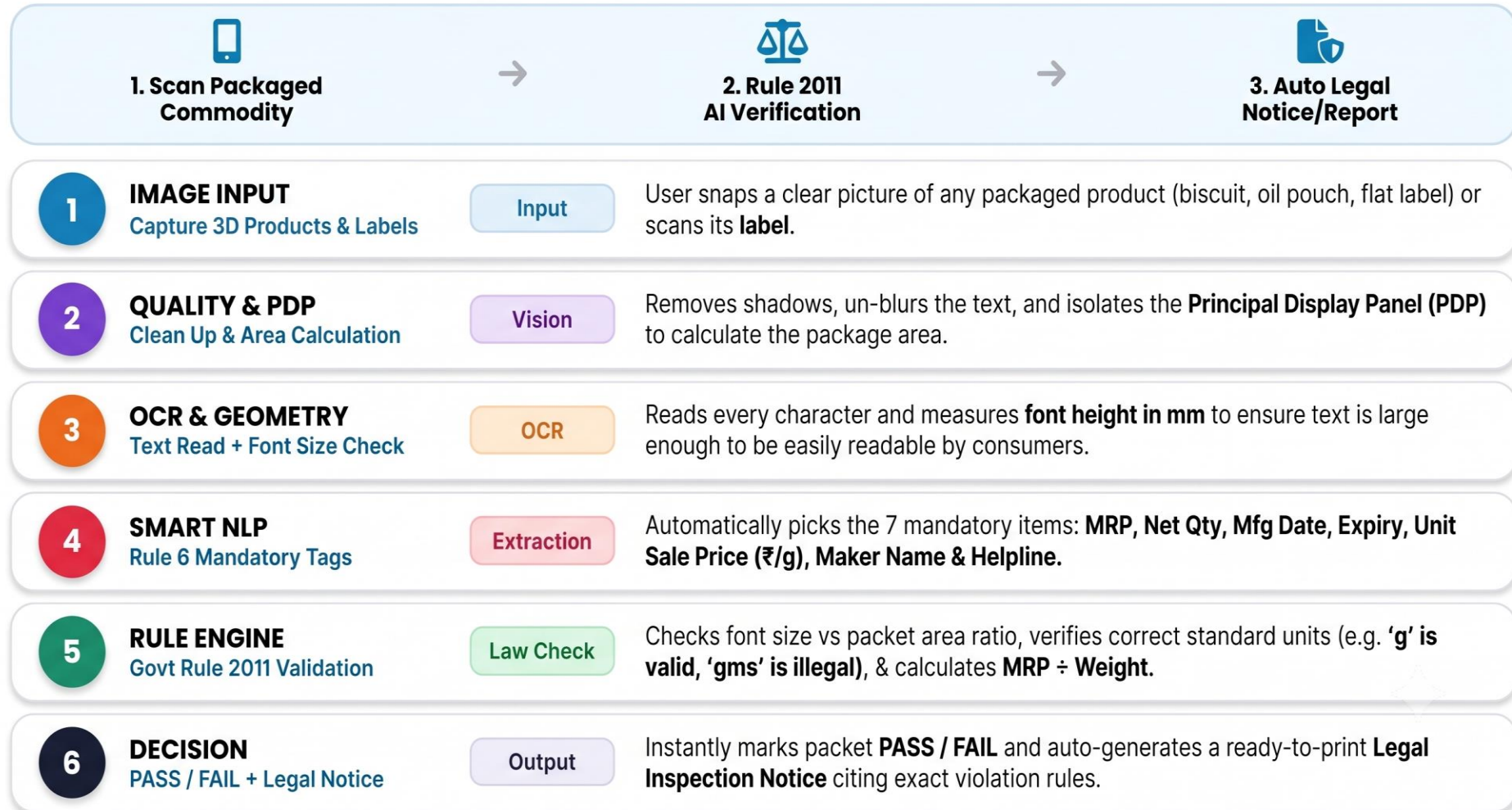
CV Vision / OCR
OpenCV + OCR engine

NLP NLP / Rules
Regex + NLP + rule engine

DB Database
PostgreSQL + object storage

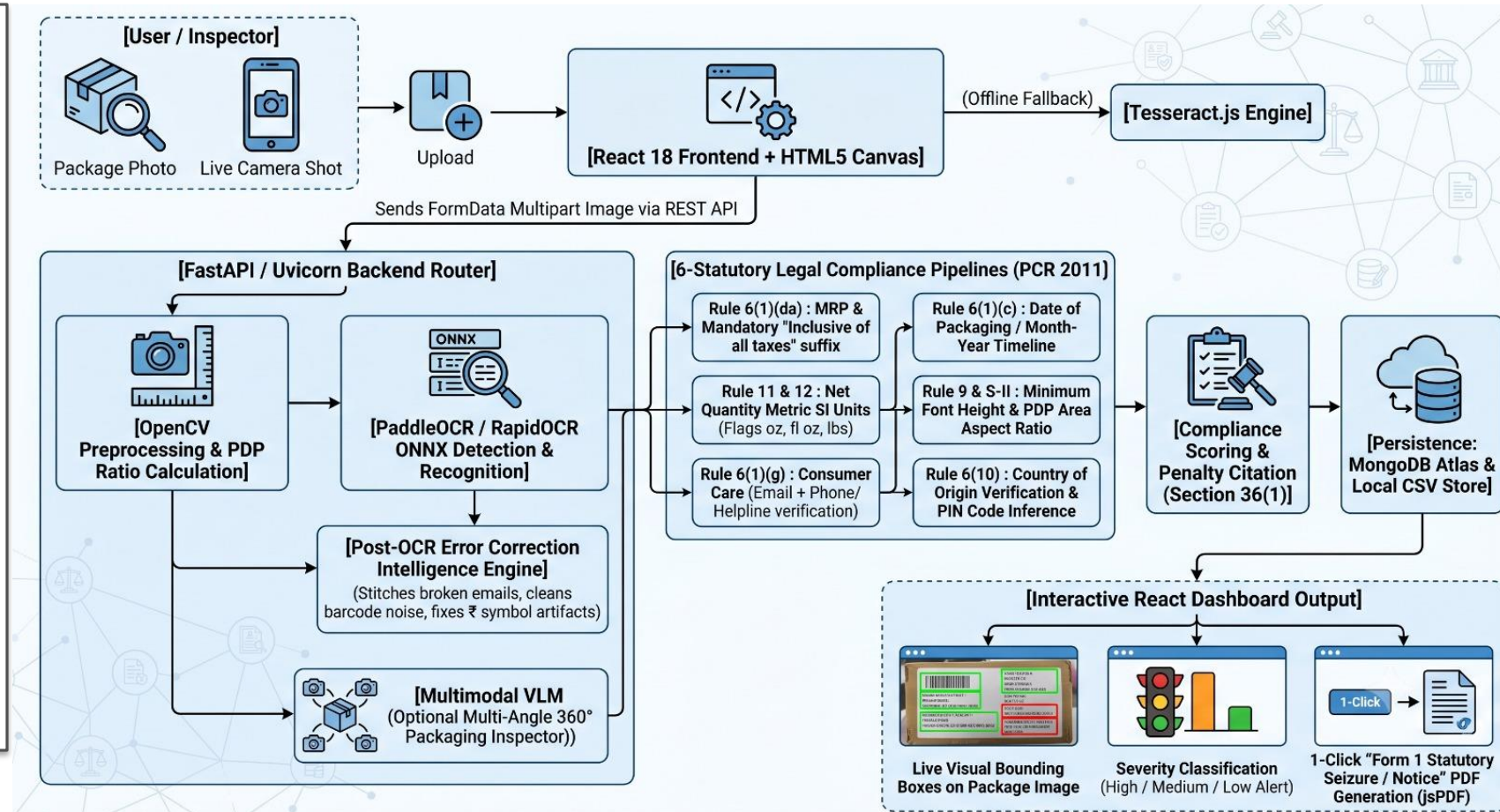
SEC Security
RBAC + audit trail

AI/ML + RULE-ENGINE WORKFLOW (HOW IT WORKS)



FEASIBILITY OF IDEA

- Computer vision and OCR are mature enough for a working prototype.
- Rule-based validation is deterministic, testable and easier to audit.
- System can start with image upload / mobile capture before deeper integrations.
- Human-in-the-loop keeps final inspection decisions under authorized users.
- Reports and dashboards fit existing digital inspection workflows.



PILOT → VALIDATE → SCALE

IMPACT AND BENEFITS

Target audience: Legal Metrology officers, consumers, e-commerce platforms, manufacturers and packers

Social

- Consumers protected from wrong MRP, short weight and missing manufacturer details
- Anyone can verify a pack before buying, straight from a phone

Economic

- Manual inspection effort and cost fall sharply
- Faster enforcement cuts losses for consumers and honest sellers
- Manufacturers get a self-check tool before printing labels

Governance

- Every scan is logged, giving objective and auditable evidence
- Dashboards reveal which categories and brands offend most often
- Enforcement shifts from random sampling to data-driven targeting

Scale and reach

- Screens lakhs of physical and online products, which manual inspection cannot match
- One rule engine deploys across every state
- Extends naturally to FSSAI and other labelling checks

RESEARCH & REFERENCES

- **Department of Consumer Affairs: Legal Metrology Act / Rules and amendments.**
- Department of Consumer Affairs, Government of India, *Legal Metrology (Packaged Commodities) Rules, 2011*. [<https://consumeraffairs.gov.in/pages/legal-metrology-act>]
- Government of India, *The Legal Metrology Act, 2009*, Department of Consumer Affairs. [<https://consumeraffairs.gov.in/pages/legal-metrology-act>]
- Scene text detection and recognition with advances in deep learning: a survey.[DOI: 10.1007/s10032-019-00320-5]
- From pixels to text: A deep learning survey of scene text detection and recognition.[DOI: 10.1016/j.compeleceng.2026.111139]

FUTURE SCOPE

- **Multilingual AI-OCR** — Support Hindi and regional Indian languages for wider product coverage.
- **Real-Time Mobile Inspection** — Enable inspectors to scan products instantly using a smartphone camera.
- **Dynamic Compliance Engine** — Automatically adapt to updates and amendments in Legal Metrology rules.
- **Government & E-Commerce Integration** — automated violation reporting and centralized dashboards.


PACKAGE INGESTION & OCR INPUT

Multi-Image Scan

Manual Text Input

 Engine:

 RapidOCR...

 Script:

 Auto-Detect...

Multi-Angle Package Ingestion (1 to 4 Photos)

1 / 4 Angles Captured

 Clear All

 Add Angle Photo

Angle 1



WhatsApp Image... 372.5 KB

Angle 2 (Back Panel)

Statutory Declarations & Mfg Info

Click or Drag Photo

Angle 3 (Side / MRP / Care)

MRP, Tax Clause & Consumer Redressal

Click or Drag Photo

Angle 4 (Mfg / Barcode)

Batch No, Barcode & Dimensions

Click or Drag Photo

 Multi-Image Aggregated OCR Active

Min 1 - Max 4 Angles

 Run Statutory Compliance Audit (1 Angles)

CORE PIPELINE

IMAGE

OCR

NLP

RULES

EVIDENCE

REPORT